COs and POs Mapping

POs	Competency	Performance Indicator	CO1	CO2	CO3	CO4	CO5
PO1 Engineering Knowledge	1.1 Demonstrate competence in mathematical modeling.	1.1.1 Apply the knowledge of discrete structures, linear algebra, statistics, numerical techniques and theoretical computer science to solve problems	Yes	Yes	Yes	Yes	Yes
		1.1.2 Apply the concepts of probability, statistics and queuing theory in modeling of computer based system, data and network protocols.	Yes	Yes	Yes	Yes	Yes
	1.2 Demonstrate competence in basic sciences	1.2.1 Apply laws of natural science to an engineering problem					
	1.3 Demonstrate competence in engineering fundamentals	1.3.1 Apply engineering fundamentals	Yes	Yes	Yes	Yes	Yes
	1.4 Demonstrate competence in specialized engineering knowledge to the program	1.4.1 Apply theory and principles of computer science and information technology to solve an engineering problem.	Yes	Yes	Yes	Yes	Yes
Total No of PI : 5		**counting the Number of PI attained**	4	4	4	4	4
No of PI Attained / Total of PI's * 100		**% of PI attained**	80	80	80	80	80
Level 1: % of PI%=59% Level 2: 60<=% of PI< =69 Level 3:% of PI >= 70		**Level**	3	3	3	3	3

POs	Competency	Performance Indicator	CO1	CO2	CO3	CO4	CO5
PO 2 Problem Analysis	2.1 Demonstrate an ability to identify and formulate complex engineering problems	2.1.1 Articulate problem statements and identify objectives	Yes	Yes	Yes	Yes	Yes
		2.1.2 Identify engineering systems, variables, and parameters to solve problems	Yes	Yes	Yes	Yes	Yes
		2.1.3 Identify the mathematical, engineering and other relevant knowledge that applies to a given problem	Yes	Yes	Yes	Yes	Yes
	2.2 Demonstrate an ability to formulate a solution plan and methodology	2.2.1 Reframe complex problems into interconnected sub-problems		YES	YES	YES	YES
		2.2.2 Identify, assemble and evaluate information and resources					
		2.2.3 Identify existing processes/solution methods for solving the problem, including forming justified approximations and assumptions	YES	YES	YES	YES	YES
		2.2.4 Compare and contrast alternative solution processes to select the best process					YES
	2.3 Demonstrate an ability to formulate and interpret a model	2.3.1 Combine scientific principles and engineering concepts to	YES	YES	YES	YES	YES

		formulate models (mathematical or otherwise) of a system or process					
		2.3.2 Identify assumptions necessary to allow modeling at the required level of accuracy					
	2.4 Demonstrate an ability to execute a solution process and analyze results	2.4.1 Apply engineering mathematics and computations to solve mathematical models	YES	YES	YES	YES	YES
		2.4.2 Produce and validate results through skilful use of contemporary engineering tools	YES	YES	YES	YES	YES
		2.4.3 Identify sources of error in the solution process and limitations of the solution	NO	YES	NO	YES	YES
		2.4.4 Extract desired understanding and conclusions consistent with objectives and limitations	YES	YES	YES	YES	YES
Total No of PI : 13		**counting the Number of PI attained**	8	10	9	10	11
No of PI Attained / Total of PI's * 100		**% of PI attained**	61.54	76.92	69.23	76.92	84.62
Level 1: % of PI%=59% Level 2: 60<=% of PI< =69 Level 3:% of PI >= 70		**Level**	2	3	2	3	3

POs	Competency	Performance Indicator	CO1	CO2	CO3	CO4	CO5
PO3 Design /Development of Solutions	3.1 Demonstrate an ability to define a complex/open-ended problem in engineering terms	3.1.1 Recognize that need analysis is key to good problem definition	YES	YES	YES	YES	YES
		3.1.2 Elicit and document engineering requirements from stakeholders	YES	YES	YES	YES	YES
		3.1.3 Synthesize engineering requirements from a review of the state-of-the-art	YES	YES	YES	YES	YES
		3.1.4 Extract engineering requirements from relevant engineering codes and standards	YES	YES	YES	YES	YES
		3.1.5 Explore and synthesize requirements considering health, safety, environmental, cultural and societal issues	YES	YES	YES	YES	YES
		3.1.6 Determine design objectives, functional requirements and arrive at specifications	YES	YES	YES	YES	YES
	3.2 Demonstrate an ability to generate a diverse set of alternative design solutions	3.2.1 Apply formal idea generation tools to develop multiple engineering design solutions	YES	YES	YES	YES	YES
		3.2.2 Build models/prototypes/algorithms to develop a diverse set of design solutions					
		3.2.3 Identify suitable criteria for the evaluation of alternate design solutions	YES	YES	YES	YES	YES
	3.3 Demonstrate an ability to select optimal design scheme for further development	3.3.1 Apply formal decision-making tools to select optimal engineering design solutions	YES	YES	YES	YES	YES
		3.3.2 Consult with domain experts and stakeholders to select candidate design solution for further development	YES	YES	YES	YES	YES
	3.4 Demonstrate an	3.4.1 Refine a conceptual design into a detailed	YES	YES	YES	YES	YES

	ability to advance an engineering design to defined end state	design within the existing constraints					
		3.4.2 Generate information through appropriate tests to improve or revise the design	YES	YES	YES	YES	YES
Total No of PI : 13		counting the Number of PI attained	8	8	8	8	8
No of PI Attained / Total of PI's * 100		% of PI attained	61.54	61.54	61.54	61.54	61.54
Level 1: % of PI%=59% Level 2: 60<=% of PI< =69 Level 3:% of PI >= 70		Level	2	2	2	2	2

POs	**Competency**	**Performance Indicator**	**CO1**	**CO2**	**CO3**	**CO4**	**CO5**
PO 4 Conduct Investigations of Complex Problems	4.1 Demonstrate an ability to conduct investigations of technical issues consistent with their level of knowledge	4.1.1 Define a problem, its scope and importance for purposes of investigation	YES	YES	YES	YES	YES
		4.1.2 Examine the relevant methods, tools and techniques of experiment design, system calibration, data acquisition, analysis and presentation					
		4.1.3 Apply appropriate instrumentation and/or software tools to make measurements					
		4.1.4 Establish a relationship between measured data and underlying principles	YES	YES	YES	YES	YES
	4.2 Demonstrate an ability to design experiments to solve open-ended problems	4.2.1 Design and develop an experimental approach, specify appropriate equipment and procedures	YES	YES	YES	YES	YES

		4.2.2 Understand the importance of statistical design of experiments and choose an appropriate experimental design plan				YES	
	4.3 Demonstrate an ability to analyze data and reach valid conclusions	4.3.1 Use appropriate procedures, tools and techniques to conduct experiments and collect data	YES	YES	YES	YES	YES
		4.3.2 Analyze data for trends and correlations, stating possible errors and limitations					
		4.3.3 Represent data (in tabular and/or graphical forms) to facilitate analysis and explanation					
		4.3.4 Synthesize information and knowledge from raw data to reach appropriate conclusions					
Total No of PI : 10		counting the Number of PI attained	4	4	4	5	4
No of PI Attained / Total of PI's * 100		% of PI attained	40	40	40	50	40
Level 1: % of PI%=59% Level 2: 60<=% of PI< =69 Level 3:% of PI >= 70		Level	1	1	1	1	1

POs	Competency	Performance Indicator	CO1	CO2	CO3	CO4	CO5
PO5:Modern tool usage	5.1 Demonstrate an ability to identify/create modern engineering tools, techniques and resources	5.1.1 Identify modern engineering tools such as AI/ML frameworks, data analytics platforms, modeling tools, techniques and resources	YES	YES	YES	YES	YES

		5.1.2 Create/adapt/modify/extend tools and techniques to solve engineering problems	YES	YES	YES	YES	YES
	5.2 Demonstrate an ability to select and apply discipline-specific tools, techniques and resources	5.2.1 Identify the strengths and limitations of tools for (i) acquiring information, (ii) modeling and simulating, (iii) monitoring system performance, and (iv) creating engineering designs	YES	YES	YES	YES	YES
		5.2.2 Demonstrate proficiency in using discipline-specific tools					
	5.3 Demonstrate an ability to evaluate the suitability and limitations of tools	5.3.1 Discuss limitations and validate tools, techniques and resources	YES	YES	YES	YES	YES
		5.3.2 Verify the credibility of results from tool use with reference to accuracy, limitations, and assumptions					
Total No of PI : 6		counting the Number of PI attained	4	4	4	4	4
No of PI Attained / Total of PI's * 100		% of PI attained	66.67	66.67	66.67	66.67	66.67
Level 1: % of PI%=59% **Level 2: 60<=% of PI< =69** **Level 3:% of PI >= 70**		Level	2	2	2	2	2

POs	Competency	Performance Indicator	CO1	CO2	CO3	CO4	CO5
PO 6 The Engineer and the World	6.1 Demonstrate an ability to describe engineering roles in a broader context pertaining to environment, health, safety, legal and public welfare	6.1.1 Identify and describe various engineering roles particularly as pertains to protection of the public and public interest at global, regional and local levels	NO	NO	YES	YES	YES
	6.2 Demonstrate an understanding of professional engineering regulations, legislation and standards	6.2.1 Interpret legislation, regulations, codes, and standards relevant to the discipline and explain its contribution to	NO	NO	NO	NO	NO

		the protection of the public					
Total No of PI : 2		counting the Number of PI attained	0	0	1	1	1
No of PI Attained / Total of PI's * 100		% of PI attained	0	0	50%	50%	50%
Level 1: % of PI%=59% Level 2: 60<=% of PI< =69 Level 3:% of PI >= 70		Level	0	0	1	1	1
PO7: Environment and sustainability	7.1 Demonstrate an understanding of the impact of engineering and industrial practices on social, environmental and economic contexts	7.1.1 Identify risks/impacts in the life-cycle of an engineering product or activity	NO	NO	NO	NO	NO
		7.1.2 Understand the relationship between technical, socio-economic and environmental dimensions of sustainability	NO	NO	NO	NO	NO
	7.2 Demonstrate an ability to apply principles of sustainable design and development	7.2.1 Describe management techniques for sustainable development	NO	NO	NO	NO	NO
		7.2.2 Apply principles of preventive engineering and sustainable development to engineering activities or products	NO	NO	NO	NO	NO
Total No of PI : 4		counting the Number of PI attained	0	0	0	0	0
No of PI Attained / Total of PI's * 100		% of PI attained	0	0	0	0	0
Level 1: % of PI%=59% Level 2: 60<=% of PI< =69 Level 3:% of PI >= 70		Level	-	-	-	-	-

POs	**Competency**	**Performance Indicator**	**CO1**	**CO2**	**CO3**	**CO4**	**CO5**

PO8: Ethics	8.1 Demonstrate an ability to recognize ethical dilemmas	8.1.1 Identify situations of unethical professional conduct and propose ethical alternatives	YES	YES	YES	YES	YES
	8.2 Demonstrate an ability to apply the Code of Ethics	8.2.1 Identify tenets of professional code of ethics	NO	NO	NO	NO	NO
		8.2.2 Examine and apply moral & ethical principles to known case studies	NO	NO	NO	NO	NO
		8.2.3 Demonstrate commitment to human values, diversity and inclusion	NO	NO	NO	NO	NO
Total No of PI : 4		counting the Number of PI attained	1	1	1	1	1
No of PI Attained / Total of PI's * 100		% of PI attained	25	25	25	25	25
Level 1: % of PI%=59% Level 2: 60<=% of PI< =69 Level 3:% of PI >= 70		Level	1	1	1	1	1

POs	**Competency**	**Performance Indicator**	**CO1**	**CO2**	**CO3**	**CO4**	**CO5**
PO9: Individual and team work	9.1 Demonstrate an ability to form a team and define a role for each member	9.1.1 Recognize a variety of working and learning preferences; appreciate the value of diversity on a team	YES	YES	YES	YES	YES
		9.1.2 Implement the norms of practice (e.g. rules, roles, charters, agendas, etc.) of effective team work to					

		accomplish a goal					
	9.2 Demonstrate effective individual and team operations—communication, problem-solving, conflict resolution and leadership skills	9.2.1 Demonstrate effective communication, problem-solving, conflict resolution and leadership skills	YES	YES	YES	YES	YES
		9.2.2 Treat other team members respectfully	YES	YES	YES	YES	YES
		9.2.3 Listen to other members	YES	YES	YES	YES	YES
		9.2.4 Maintain composure in difficult situations	YES	YES	YES	YES	YES
	9.3 Demonstrate success in a team-based project	9.3.1 Present results as a team, with smooth integration of contributions from all individual efforts	YES	YES	YES	YES	YES
Total No of PI : 7		counting the Number of PI attained	5	5	5	5	5
No of PI Attained / Total of PI's * 100		% of PI attained	71.42857	71.4286	71.4286	71.4286	71.43
Level 1: % of PI%=59% Level 2: 60<=% of PI< =69 Level 3:% of PI >= 75		Level	2	2	2	2	2

POs	**Competency**	**Performance Indicator**	**CO1**	**CO2**	**CO3**	**CO4**	**CO5**
P10: Communication:	10.1 Demonstrate an ability to comprehend technical literature and document project work	10.1.1 Read, understand and interpret technical and non-technical information	YES	YES	YES	YES	YES
		10.1.2 Produce clear, well-constructed, and well-supported written engineering documents					

		10.1.3 Create flow in a document or presentation - a logical progression of ideas so that the main point is clear	YES	YES	YES	YES	YES
	10.2 Demonstrate competence in listening, speaking, and presentation	10.2.1 Listen to and comprehend information, instructions, and viewpoints of others	YES	YES	YES	YES	YES
		10.2.2 Deliver effective oral presentations to technical and non-technical audiences	YES	YES	YES	YES	YES
	10.3 Demonstrate the ability to integrate different modes of communication	10.3.1 Create engineering-standard figures, reports and drawings to complement writing and presentations	NO	NO	NO	NO	NO
		10.3.2 Use a variety of media effectively to convey a message in a document or presentation	NO	NO	NO	NO	NO
		10.3.3 Communicate inclusively considering cultural, language and learning differences	YES	YES	YES	YES	YES
Total No of PI : 8		counting the Number of PI attained	5	5	5	5	5
No of PI Attained / Total of PI's * 100		% of PI attained	62.5	62.5	62.5	62.5	62.5
Level 1: % of PI%=59% Level 2: 60<=% of PI< =69 Level 3:% of PI >= 70		Level	2	2	2	2	2

POs	Competency	Performance Indicator	CO1	CO2	CO3	CO4	CO5

PO11: Project management and finance	11.1 Demonstrate an ability to evaluate the economic and financial performance of an engineering activity	11.1.1 Describe various economic and financial costs/benefits of an engineering activity					
		11.1.2 Analyze different forms of financial statements to evaluate the financial status of an engineering project					
	11.2 Demonstrate an ability to compare and contrast the costs/benefits of alternate proposals	11.2.1 Analyze and select the most appropriate proposal based on economic and financial considerations					
	11.3 Demonstrate an ability to plan/manage an engineering activity within time and budget constraints	11.3.1 Identify the tasks required to complete an engineering activity and the resources required to complete the tasks					
		11.3.2 Use project management tools to schedule an engineering project so it is completed on time and on budget					
Total No of PI : 5		counting the Number of PI attained					
No of PI Attained / Total of PI's * 100		% of PI attained					
Level 1: % of PI%=59% Level 2: 60<=% of PI< =69 Level 3:% of PI >= 70		Level					

POs	**Competency**	**Performance Indicator**	**CO1**	**CO2**	**CO3**	**CO4**	**CO5**
PO 12: Life-Long Learning	12.1 Demonstrate an ability to identify	12.1.1 Describe the rationale for the requirement	YES	YES	YES	YES	YES

	gaps in knowledge and a strategy to close these gaps	for continuing professional development					
		12.1.2 Identify deficiencies or gaps in knowledge and demonstrate an ability to source information to close this gap	YES	YES	YES	YES	YES
	12.2 Demonstrate an ability to identify changing trends in engineering knowledge and practice	12.2.1 Identify historic points of technological advance in engineering that required practitioners to seek education to stay current	YES	YES	YES	YES	YES
		12.2.2 Recognize the need and be able to clearly explain why it is vitally important to keep current regarding new developments in the field	YES	YES	YES	YES	YES
	12.3 Demonstrate an ability to identify and access sources for new information	12.3.1 Source and comprehend technical literature and other credible sources of information					
		12.3.2 Analyze sourced technical and popular information for feasibility, viability, sustainability, etc.					
Total No of PI : 6		counting the Number of PI attained	4	4	4	4	4
No of PI Attained / Total of PI's * 100		% of PI attained	66.67	66.67	66.67	66.67	66.67
Level 1: % of PI%=59% Level 2: 60<=% of PI< =69 Level 3:% of PI >= 70		Level	2	2	2	2	2

POs	Competency	Performance Indicator	CO1	CO2	CO3	CO4	CO5
PSO 1 AI Graduates should be able to evolve AI based efficient domain specific processes for effective decision making	To **develop, adapt, and apply AI-based domain-specific processes** to enhance efficiency and support **effective, data-driven decision making.**	**Design and implement AI-based, domain-specific process models**	YES	YES	YES	YES	YES
		Apply appropriate AI techniques (e.g., ML, analytics, optimization) to improve efficiency and effectiveness of domain-specific processes.	YES	YES	YES	YES	YES
		Evaluate AI-driven solutions using relevant performance metrics to justify decisions and recommendations.	NO	YES	NO	YES	YES
Total No of PI : 3		counting the Number of PI attained	2	3	2	3	3
No of PI Attained / Total of PI's * 100		% of PI attained	66.67	100.00	66.67	100.00	100.00
Level 1: % of PI%=59% Level 2: 60<=% of PI< =69 Level 3:% of PI >= 70		Level	2	3	2	3	3

POs	Competency	Performance Indicator	CO1	CO2	CO3	CO4	CO5
PSO2 Graduates should be able to arrive at actionable foresight, insight, hindsight from data for solving business and engineering problems	**To extract hindsight, insight, and foresight from data using analytical and AI-driven techniques to solve complex business and engineering problems and support informed decision making.**	**Apply descriptive analytics to derive hindsight from historical data.**	YES	YES	YES	YES	YES
		Use diagnostic and predictive analytics to generate insight into patterns, trends, and root causes.		YES	YES	YES	
		Develop predictive and prescriptive models to produce foresight for future-oriented decisions.	YES	YES	YES	YES	YES
		Translate analytical results into actionable					

		recommendations addressing real-world business and engineering challenges.					
		Validate decisions using **appropriate metrics, visualizations, and domain constraints**.	YES	YES	YES	YES	YES
Total No of PI : 5		counting the Number of PI attained	3	4	4	4	3
No of PI Attained / Total of PI's * 100		% of PI attained	60	80	80	80	60
Level 1: % of PI%=59% Level 2: 60<=% of PI< =69 Level 3:% of PI >= 70		Level	2	3	3	3	2

POs	**Competency**	**Performance Indicator**	**CO1**	**CO2**	**CO3**	**CO4**	**CO5**
PSO3 Graduates should be able to create, select and apply theoretical knowledge of AI and Data Analytics along with practical industrial tools to manage and solve wicked	**To create, select, and apply AI and Data Analytics theories and industrial tools to analyze, manage, and solve wicked societal problems in a responsible and sustainable manner.**	Identify and **formulate wicked societal problems** considering technical, social, ethical, and environmental dimensions.					
		Select and apply **appropriate AI and Data Analytics theories** to address problem complexity and uncertainty.	YES	YES	YES	YES	YES
		Use **industry-standard tools and platforms** (e.g., analytics frameworks, ML platforms, visualization tools) for solution development.	YES	YES	YES	YES	YES
		Design and implement **data-driven models** that integrate multiple stakeholder perspectives.	YES	YES	YES	YES	YES

societal problems		Evaluate solutions for **effectiveness, scalability, ethical impact, and sustainability**.	YES	YES	YES	YES	YES
Total No of PI : 6		counting the Number of PI attained	4	4	4	4	4
No of PI Attained / Total of PI's * 100		% of PI attained	80	80	80	80	80
Level 1: % of PI%=59% Level 2: 60<=% of PI< =69 Level 3:% of PI >= 70		Level	3	3	3	3	3

PANIMALAR ENGINEERING COLLEGE

An Autonomous Institution

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CO–PO–PSO Mapping for Natural Language Processing course

CO/PO/ PSO Mapping	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO1	PSO2	PSO3
CO1	3	2	2	1	2			1	2	2		2	2	2	3
CO2	3	3	2	1	2			1	2	2		2	3	3	3
CO3	3	2	2	1	2	1		1	2	2		2	2	3	3
CO4	3	3	2	1	2	1		1	2	2		2	3	3	3
CO5	3	3	2	1	2	1		1	2	2		2	3	2	3
AVG	3	2.6	2	1	2	1		1	2	2		2	2.6	2.6	3

3	Strongly Mapped	2	Moderately Mapped	1	Slightly Mapped
		Blank – Not Mapped			